The Relative Wage and Marriage after 2010: A Comment on Shenhav (2021)

Mahla Shourian* and Pallab Ghosh†

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Abstract

Shenhav (2021) reports that growth in women's relative potential earnings reduced U.S. marriage rates between 1980 and 2010. Rebuilding the analysis from raw IPUMS microdata, we reproduce published estimates and extend the sample through 2024. The extended coefficient is zero. What changed is the measure, not behavior. Within local marriage markets, correlation between the shift-share relative wage and observed relative wages collapses from $+0.744$ (1980–2010) to -0.026 (2015–2024). Rebasing shares to 1980 does not restore it. A fixed-share regressor can lose correlation while maintaining narrow standard errors. We suggest checking measure relevance period by period.

Keywords: gender wage gap, marriage, spouse quality, shift-share measure, replication

JEL codes: J12, J16, J31, C26.

*Corresponding author: Department of Economics, University of Oklahoma. 308 Cate Center Drive, Room 336 Cate 1 (CCD1), Norman, OK 73072. Email: mahla.shourian@ou.edu.

†Department of Economics, University of Oklahoma. 308 Cate Center Drive, Room 323 Cate 1 (CCD1), Norman, OK 73072. Email: pallab.ghosh@ou.edu.

I Introduction

In a recent paper in this Review, Shenhav (2021) shows that growth in women's earnings opportunities relative to men's reduced marriage in the United States. The analysis draws on Census and American Community Survey (ACS) data from 1980 through 2010, and it relates the marital status of women in an education, race, and state marriage market to a 1970 weighted shift share measure of the female-to-male potential wage. The estimated effect is large. A 10 percent increase in the relative potential wage lowers the share of women who are married by 5.1 percentage points, and it raises the shares who are divorced and who have never married.

The result speaks to a central prediction of marriage market models. Where marriage delivers gains from specialization, a woman whose own earnings opportunities improve has less to gain from marrying, and marriage rates should fall (Becker, 1973). Because the retreat from marriage among American women is one of the largest demographic changes of the past four decades, evidence that the narrowing gender wage gap accounts for part of it carries considerable weight.

The finding also raises a question. The estimates are identified from data that end in 2010, and the marriage market has changed a great deal since then. Marriage kept falling; the relative potential wage did not keep rising. Figure 1 shows both series together. Through 2010 the relative wage climbed steadily, from 0.77 to 0.83, while the married share fell and the share never married rose. After 2010 the relative wage is flat, ending at 0.82 in 2024, and yet marriage continues to decline. Rebuilding the analysis from raw IPUMS microdata, our estimate of the marriage coefficient over 1980 to 2010 is -0.0502, against a published -0.0513. In a window built only from the new waves, 2015 to 2024, it is -0.0049 with a standard error of 0.0095.

Three things could account for the smaller coefficient. First, women may simply respond differently now than they did before. Second, the constructed measure may have stopped tracking the wage women face, so that the coefficient is attenuated toward zero whatever behavior does. Third, the shorter window may not have the power to detect an effect that is still there. We find that the second explanation is the operative one.

The main work of this comment establishes why that number cannot be read as evidence about behavior. Within a marriage market, the correlation between the constructed relative wage and the wage women face falls from $+0.744$ over the original period to -0.026 over 2015 to 2024. Holding behavior fixed at the value implied by the original period, the collapse of the measure alone predicts a coefficient of $+0.0015$ after 2010. We observe -0.0045 with a standard error of 0.0099. The two differ by 0.60 standard errors, so every estimate we report for the later waves is consistent with women responding to the relative wage exactly as they always did.

What follows is therefore a statement about the design rather than about behavior. The estimate in Shenhav (2021) is identified by the rapid and geographically uneven closing of the gender wage gap between 1980 and 2010. That episode is over. After 2010 the relative potential wage is nearly constant within marriage markets, and the shift share construction no longer resolves what movement remains. One qualification applies to our reconstruction throughout: because the occupation crosswalk and top coding rules derive from intermediates supplied by the author, our rebuild begins from raw microdata but retains the author's underlying classification pipeline. Section II describes the estimating equation and data. Section III reports the reproduction and extension. Section IV shows why the extended estimates are uninformative. Section V concludes. An online appendix collects secondary outcomes, robustness checks, and supporting exhibits.

II Specification and Data

II.1 The estimating equation

The unit of observation is a marriage market defined by education, race, and state, divided further by birth cohort and survey year. Following Shenhav (2021), we estimate

$$Y_{\mu ct} = \beta \log(\text{RelWage}_{\mu t}) + \alpha_\mu + \delta_{rt} + \chi_{et} + \gamma_{st} + \xi_{ct} + \rho_{rs} t + X'_{\mu t}\phi + \varepsilon_{\mu ct}, \tag{1}$$

where μ indexes marriage markets, c birth cohorts, and t survey years. The fixed effects absorb permanent differences across marriage markets (α_μ) together with year shocks specific to a race (δ_{rt}), an education group (χ_{et}), a state (γ_{st}), and a birth cohort (ξ_{ct}), and $\rho_{rs}t$ allows a linear trend for each race and state pair. The controls $X_{\mu t}$ are the mean years of schooling of women and of men and the sex ratio in the market. The outcomes $Y_{\mu ct}$ are the shares of women in a cell who are married, divorced, or never married. Cells are weighted by their female population, and standard errors are clustered by state. As in the original study, coefficients are scaled to a 10 percent increase in the relative potential wage. A second specification adds the average log potential wage, so that β measures the effect of the relative wage holding the wage level fixed.

The potential wage is a weighted average of national wage rates across cells defined by industry and occupation, computed leaving out the own state, with weights fixed at 1970 employment shares within each marriage market (Bartik, 1991; Aizer, 2010; Shenhav, 2021). Two features of this construction matter here. First, because we use the author's occupation crosswalk and the author's rules for top coding, the close agreement reported below validates the rebuilt pipeline as a whole rather than the shift-share series on its own, and we do not present it as more than that (Goldsmith-Pinkham et al., 2020). Second, the constructed measure enters equation (1) directly as a regressor. It is not the first stage of a two stage least squares procedure, so the diagnostics that accompany an instrumental variables design do not apply, and whether the regressor still measures what it is meant to measure is an empirical question about the regressor itself. Section IV treats it as one.

II.2 Data

For the original period we reconstruct the analysis from IPUMS USA microdata for the 1980, 1990, and 2000 Censuses and the 2010 ACS (Ruggles et al., 2025). The sample consists of women aged 22 to 44 in two education groups, high school or less and some college or more, and three race and ethnicity groups, non-Hispanic White, non-Hispanic Black, and Hispanic. National wage inputs come from IPUMS CPS files (Flood et al., 2025), and the fixed weights come from the 1970 Census. The extension adds five ACS waves after 2010, namely 2015, 2017, 2019, 2022, and 2024, together with CPS wage data through 2025. The number of cells roughly doubles, from 23,774 to 52,184. Every step of the construction is applied identically to the new waves. We impose the age restriction at the record level in each wave, and we build the schooling and sex ratio controls directly from the contemporaneous microdata.

One difference between the periods deserves mention, because it is the kind of thing that can generate a spurious null. The 1980, 1990, and 2000 samples come from the Census long form, while every later wave comes from the ACS. The two differ in their sampling frame, their questionnaire, and their treatment of group quarters. However, this transition does not fall at the boundary between our two windows, since the 2010 wave that closes the original sample is already an ACS wave. The change in survey design therefore sits inside the period over which the published relationship holds, which is the opposite of what an explanation resting on survey design would require.

III Reproduction and Extension

III.1 The original period reproduces

We begin on the author's released data. Every specification that file supports reproduces the published point estimates, standard errors, and significance levels to the precision reported in Shenhav (2021), across the main and appendix exhibits. The more demanding test rebuilds the analysis from raw microdata. The first two columns of Table 1 place our estimates beside the published ones. In Panel A the marriage coefficient is -0.0502 with a standard error of 0.0086, against a published -0.0513. The divorce coefficient is $+0.0211$ against $+0.0193$, and the coefficient for never marrying is $+0.0304$ against $+0.0323$. Panel B adds the average potential wage, and there the marriage estimate is -0.0496 against a published -0.0478. Signs, magnitudes, and significance all agree, and we treat the original results as reproduced.

III.2 The extended estimates

The coefficients fall once the sample is extended. Columns 3 and 4 of Table 1 report the extension. Column 3 adds every wave after 2010 to form a 1980 to 2024 panel, and column 4 uses only the later waves, 2015 to 2024. For marriage, the Panel A coefficient of -0.0502 falls to -0.0224 in the pooled panel and to -0.0049 (standard error 0.0095) in the later waves alone. Panel B gives the same sequence: -0.0496, then -0.0306, then -0.0104. Pooling the two windows and interacting the log relative wage with an indicator for the later waves, the change is $+0.0453$ with a standard error of 0.0137 and a p value of 0.002 (Online Appendix Table OA.5). Section IV returns to what that test measures.

The pooled 1980 to 2024 coefficient remains negative and significant at -0.0224, and we want to be clear about what it does and does not show. A panel that averages three decades in which the coefficient was large against one decade in which it is zero will return a significant pooled estimate whether or not anything survives at the end. The informative comparison is between periods. Divorce and never marrying attenuate in the same way. The divorce coefficient falls from $+0.0211$ over 1980 to 2010 to -0.0008 over the full panel and -0.0076 in the later waves. The coefficient for never marrying falls from $+0.0304$ to $+0.0235$ in the full panel and $+0.0122$ in the later waves. Over the same interval the married share among women aged 22 to 44 falls from 0.651 to 0.518, and the share never married rises from 0.238 to 0.402.[1]

IV Why the Extended Estimates Are Uninformative

A null of this kind invites a mechanical explanation, and the obvious one is that the 1970 employment weights have simply gone stale. We test this directly. Therefore, we ask whether the constructed relative wage still tracks the relative wage women actually face within a marriage mar-

[1] The secondary outcomes are extended to 2024 in online appendix Section OA.4, and the controls measured at the market level are examined in Section OA.3. Those controls shift the coefficients modestly, but they never change a sign or a significance level in any window. All of the estimates this comment reports for the later waves are subject to the argument of Section IV.

ket. Equation (1) contains marriage market fixed effects, so every permanent difference across markets is removed before the coefficient is formed, and what identifies β is movement within a market over time. Table 2 reports the answer for each of the three series that enter the construction.

Both wage levels survive. The female potential wage is tracked at $+0.991$ over 1980 to 2010 and $+0.973$ over 2015 to 2024, and the male potential wage at $+0.990$ and $+0.973$, which rules out a processing error or an artifact of the top codes. The relative wage is a different matter. Its correlation falls from $+0.744$ to -0.026. The constructed relative wage is a small residual between two large and highly correlated series, so it inherits little signal and a great deal of noise, and after 2010 the signal is gone.

IV.1 The constructed measure and the wage women face

Table 2 also shows why. Within a marriage market the standard deviation of the observed relative wage falls from 0.0702 to 0.0323, and that of the shift-share measure from 0.0273 to 0.0127. Both roughly halve. This is the same fact that Figure 1 reports at the national level, where the relative wage climbs through 2010 and is then flat, except that it holds within marriage markets as well, and what little movement remains in the constructed measure no longer lines up with what little movement remains in the data. Identification in Shenhav (2021) rests on the rapid and geographically uneven closing of the gender wage gap between 1980 and 2010. That episode is over.

IV.2 It is not the fixed effects

A natural objection is that the fixed effects in equation (1) absorb the variation we are looking for, and that the problem is the specification rather than the measure. Table 3 addresses this. We estimate four specifications, from the full baseline down to market and year fixed effects alone, and at each rung we run the measurement check alongside the marriage regression.

The objection does not survive. Removing the state by year fixed effects does not revive the check, and neither does removing the trends. At rung four, where the only fixed effects are marriage market and year, the check is -0.012 with a standard error of 0.075, so the 95 percent interval $[-0.159, +0.135]$ excludes the value of $+0.380$ that the same specification delivers over 1980 to 2010. This is a well measured zero and not an underpowered one. A second piece of evidence points the same way. If absorption were the story, the share of the constructed measure's variance surviving the projection would be much smaller in the later waves. It is 0.9 percent over 1980 to 2010 and 1.2 percent over 2015 to 2024.

IV.3 What the collapse implies for the reported coefficient

Let $X_{\mu t}$ be the relative wage women actually face in market μ at time t, and let $Z_{\mu t}$ be the constructed measure, both taken as deviations from the market mean, which is what equation (1) leaves after the fixed effects. Suppose behavior is $Y_{\mu t} = \beta^* X_{\mu t} + \varepsilon_{\mu t}$ with $E[\varepsilon_{\mu t} \mid Z_{\mu t}] = 0$. We do not regress Y on X. We regress Y on Z, so the estimator converges to

$$\hat{\beta}_Z \xrightarrow{p} \beta^* \frac{\mathrm{Cov}(Z,X)}{\mathrm{Var}(Z)} = \beta^* \rho_{ZX} \frac{\sigma_X}{\sigma_Z} \equiv \beta^* \kappa, \tag{2}$$

where ρ_{ZX} is the within-market correlation in Table 2 and σ_X and σ_Z are the corresponding standard deviations. The factor κ is a property of the measure and not of behavior.

Two features of equation (2) make this more than an inconvenience. As $\rho_{ZX} \to 0$, the estimate goes to zero whatever β^* is, so a researcher who does not run the check sees a small coefficient and reads it as a small effect. And the standard error gives no warning, because the precision of $\hat{\beta}_Z$ is governed by σ_Z and the residual variance rather than by ρ_{ZX}. A measure can be pure noise with respect to X and still deliver a tight standard error. This is the opposite of the familiar case of a weak instrument, where the standard error grows and the problem announces itself. Here the estimate looks clean and precise while carrying no information, and our -0.0045 with a standard error of 0.0099 has exactly that shape.

Substituting from Table 2 gives $\kappa = +1.913$ over 1980 to 2010 and $\kappa = -0.066$ over 2015 to 2024. Our estimate for the original period is -0.0428, so inverting equation (2) implies a behavioral parameter of $\beta^* = -0.0224$. Holding behavior fixed at that value and asking what should be observed in the later waves given only the change in the measure gives $\hat{\beta}_{Z,\text{post}} = +0.0015$. The observed value is -0.0045 with a standard error of 0.0099, which sits 0.60 standard errors away. Figure 2 reports a Monte Carlo built on the same calibration, with behavior held constant by construction. Across 4,000 draws, 95 percent of the simulated estimates fall between -0.0184 and $+0.0209$. Our observed value lies inside that interval. Our estimate from the original period lies far outside it.

This is what disarms the interaction test in Section III. The change of $+0.0453$ is a real and precisely estimated fact about the coefficient. It is not a fact about women. A regressor whose relationship to its target moves from $\rho_{ZX} = +0.744$ to $\rho_{ZX} = -0.026$ will produce a change of that kind with no change in behavior whatsoever, and equation (2) says by how much.

IV.4 Rebasing the shares does not repair the measure

If the trouble is that 1970 employment shares no longer describe where people work, then moving the base year forward should help. We built the shift-share measure again on 1980 employment shares, taken from a new extract of the 1980 Census five percent sample, and repeated everything. Table 3 reports both bases side by side. The 1980 base does what a fresher weighting scheme should do over 1980 to 2010 and nothing at all in the later waves. Within a marriage market the correlation between the observed and the constructed relative wage over 1980 to 2010 rises from 0.740 to 0.809, and the rung one check rises from 0.593 to 0.799.[2]

The correlation after 2010 is -0.007, and we find no evidence that the 1980 base restores the check in the later waves. No rung produces a positive and significant check. One result of the rebasing bears on the original study rather than on the extension. Estimated over 1980 to 2010 with contemporaneous 1980 shares, the marriage coefficient is -0.0623. The published result therefore

[2]The 0.740 reported here differs trivially from the $+0.744$ in Table 2 because the rebasing merge drops a small number of cells. Both are computed on the 1970 base and agree on every substantive quantity.

survives on 1980 shares. It is not an artifact of the base year.

IV.5 When the identifying variation ended

The path of the coefficient through time is consistent with a measure that decays rather than with a behavior that breaks. Estimated wave by wave, the marriage coefficient remains significant through 2017, becomes insignificant in 2019, and is near zero by 2024. An expanding window shows the same drift as the later waves enter (Online Appendix Section OA.5).

Two further results point the same way. Estimating equation (1) on pairs of adjacent waves inside the original sample, the 1990 and 2000 pair returns -0.0616 with a standard error of 0.0172, so a panel of two waves does not by itself destroy the result. The 2000 and 2010 pair returns -0.0016 with a standard error of 0.0120, which is the same null we report for 2015 to 2024. Correlations of decadal long differences between the constructed and the observed relative wage run between 0.12 and 0.28 over 1980 to 2010 and -0.021 over 2010 to 2024. Thus the identifying variation was already thinning inside the original sample, and whatever happened did not happen in 2010.

V Conclusion

Shenhav (2021) asks a question worth asking and answers it carefully for the period it studies. Our reproduction confirms that answer. Rebuilt from raw microdata the published relationships appear with the same signs, the same significance, and closely similar magnitudes, and they survive a change of base year for the shift-share weights. The extension does not overturn that answer so much as show that the design cannot be asked the question again. The estimate is identified by the rapid and geographically uneven closing of the gender wage gap between 1980 and 2010. After 2010 the relative potential wage is nearly constant within marriage markets, the shift-share construction no longer tracks what movement remains, and rebasing the weights to 1980 does not repair it. The coefficients we report for the later waves are consistent with women responding to the relative wage exactly as they did before, and they are equally consistent with no response at all. The published estimate therefore describes an episode rather than a parameter.

The general lesson is worth stating once, because it is not specific to this literature. A regressor constructed from fixed shares can lose its correlation with the quantity it stands in for and still deliver a narrow standard error, so the point estimate looks precise and means nothing. Nothing in the usual output signals this. We therefore suggest that a paper extending such a design over several decades report a check that the constructed series still tracks its target, computed period by period and on the variation that identifies the coefficient. Our own exercise points to one thing more. We could not rebuild the shift-share measure from public sources alone, and a measure carrying this much weight ought to be reconstructible from the public data.

Conflicts of Interest

The authors declare that they have no conflicts of interest relevant to the content of this comment, and no relevant financial or nonfinancial interests to disclose.

Data Availability Statement

This study uses IPUMS USA and IPUMS CPS microdata (Ruggles et al., 2025; Flood et al., 2025) and the replication package of Shenhav (2021). IPUMS data are available to registered users, and we disclose that registration is required in order to obtain the extracts. Raw extracts are not redistributed. Upon acceptance the authors will deposit the full replication package, including extract specifications, portable path configuration, and code for every reported analysis. The occupation crosswalk and the rules for top coding used in the reconstruction derive from intermediates supplied by the author and are included with that provenance disclosed.The controls file used throughout is `controls_ext.dta` and this file reproduces Table 1. The 1980 employment shares used in Section IV are built from a separate IPUMS USA extract of the 1980 Census five percent state sample, whose specification and build scripts are included in the package.

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Table 1: Extended Impact of the Relative Wage on Marriage, 1980 to 2024

	Author 1980–2010 (published)	Ours 1980–2010 (strict replication)	Ours 1980–2024 (extended)	Ours 2015–2024 (new only)
Panel A: Effect of 10% increase in relative wage				
Married	−0.0513***	−0.0502***	−0.0224***	−0.0049
	(0.0142)	(0.0086)	(0.0056)	(0.0095)
Divorced	+0.0193***	+0.0211***	−0.0008	−0.0076*
	(0.0061)	(0.0053)	(0.0029)	(0.0041)
Never Married	+0.0323***	+0.0304***	+0.0235***	+0.0122
	(0.0100)	(0.0074)	(0.0051)	(0.0077)
Panel B: Relative and average potential wage				
Married – rel. wage	−0.0478***	−0.0496***	−0.0306***	−0.0104
	(0.0086)	(0.0074)	(0.0057)	(0.0123)
Married – avg. wage	+0.0791***	+0.0660***	+0.0816***	+0.0214
	(0.0105)	(0.0109)	(0.0069)	(0.0183)
Divorced – rel. wage	+0.0173***	+0.0208***	+0.0006	−0.0095**
	(0.0057)	(0.0053)	(0.0030)	(0.0043)
Divorced – avg. wage	−0.0435***	−0.0361***	−0.0139***	+0.0075
	(0.0075)	(0.0075)	(0.0042)	(0.0079)
Never Married – rel. wage	+0.0311***	+0.0303***	+0.0298***	+0.0191*
	(0.0079)	(0.0071)	(0.0048)	(0.0104)
Never Married – avg. wage	−0.0272***	−0.0209*	−0.0622***	−0.0267
	(0.0095)	(0.0107)	(0.0074)	(0.0165)
Mean Y: Married	0.645	0.651	0.576	0.518
Mean Y: Divorced	0.102	0.103	0.087	0.076
Mean Y: Never Married	0.245	0.238	0.330	0.402
Observations	23,573	23,774	52,184	28,410

All ours columns use the same strict raw-IPUMS construction and published controls/specification (the detailed 1980–2010 comparison is in online appendix A). Standard errors are clustered by state. Women are restricted to record-level ages 22–44. A separate machine-readable sensitivity additionally requires the implied cohort age (*year – birthyear*) to be 22–44. Because the occupation crosswalk and historical top-code rules are recovered from author-supplied intermediates, 'raw' does not mean independent of every author mapping. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Table 2: Correlation between the Observed and the Constructed Wage, within Marriage Market

	Correlation		S.d. of the observed series	
Series	1980–2010	2015–2024	1980–2010	2015–2024
Female wage	+0.991	+0.973	0.4494	0.1395
Male wage	+0.990	+0.973	0.3880	0.1355
Relative wage	+0.744	−0.026	0.0702	0.0323

Each series is deviated from its own marriage market mean within the window, and the correlation between the observed and the constructed series is then computed with weights given by the female population of the cell. Marriage markets are defined by education, race, and state, as in equation (1). The observed series are built from the raw census and American Community Survey microdata following the construction of the original study, and the constructed series are the shift-share measures used as regressors throughout. Computed on the merged cell file underlying every estimate in this comment, all nine waves, 2,394 market by year observations. The standard deviation of the constructed relative wage, computed the same way, is 0.0273 over 1980–2010 and 0.0127 over 2015–2024.

Table 3: Measurement Check and Marriage Coefficient at Four Levels of Saturation, under 1970 and 1980 Employment Shares

		Check: observed on constructed		Share married	
Specification	Base	1980–2010	2015–2024	1980–2010	2015–2024
(1) Full equation (1)	1970	0.593***	−0.029	−0.0428***	−0.0045
		(0.191)	(0.142)	(0.0074)	(0.0099)
	1980	0.799***	−0.231	−0.0623***	+0.0031
		(0.272)	(0.317)	(0.0101)	(0.0225)
(2) Drop state by year	1970	0.619***	+0.016	−0.0412***	−0.0012
		(0.176)	(0.126)	(0.0076)	(0.0082)
	1980	0.841***	−0.052	−0.0569***	+0.0114
		(0.263)	(0.204)	(0.0094)	(0.0131)
(3) Also drop race by state trends	1970	0.723***	+0.047	−0.0381***	−0.0017
		(0.210)	(0.109)	(0.0075)	(0.0063)
	1980	1.040***	+0.118	−0.0522***	+0.0131
		(0.312)	(0.200)	(0.0081)	(0.0103)
(4) Market and year fixed effects only	1970	0.380***	−0.012	−0.1307***	−0.0026
		(0.104)	(0.075)	(0.0053)	(0.0069)
	1980	0.594***	−0.225*	−0.1670***	−0.0289***
		(0.130)	(0.124)	(0.0081)	(0.0083)
Observations, outcome	1970	23,775	28,413	23,775	28,413
	1980	25,406	30,243	25,406	30,243

The measurement check regresses the observed relative wage of a marriage market on the constructed relative wage; the marriage columns regress the share of women who are married on the same constructed measure. The two are estimated at the same level of saturation in each row, so that an outcome coefficient is read only where the regressor has been shown to carry information. Row (1) is equation (1). Row (2) removes the state by year fixed effects, row (3) also removes the race by state linear trends, and row (4) retains only marriage market and year fixed effects. Base gives the census year whose employment shares weight the shift-share construction. Standard errors in parentheses are clustered by state and cells are weighted by female population. Marriage coefficients are scaled to a 10 percent increase in the relative wage and use the specification without controls measured at the market level, which is the one that reproduces the existing exhibits exactly. Row (1) under the 1970 base reproduces the published values, including −0.0428 and −0.0045 for marriage, +0.0231 and −0.0075 for divorce, and +0.0183 and +0.0116 for never marrying. The race by state trends are materialized as regressors rather than absorbed as varying slopes; absorbing them returns −0.0387 in place of −0.0428. Cell counts differ across base years because the 1980 share matrix covers a slightly different set of markets. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

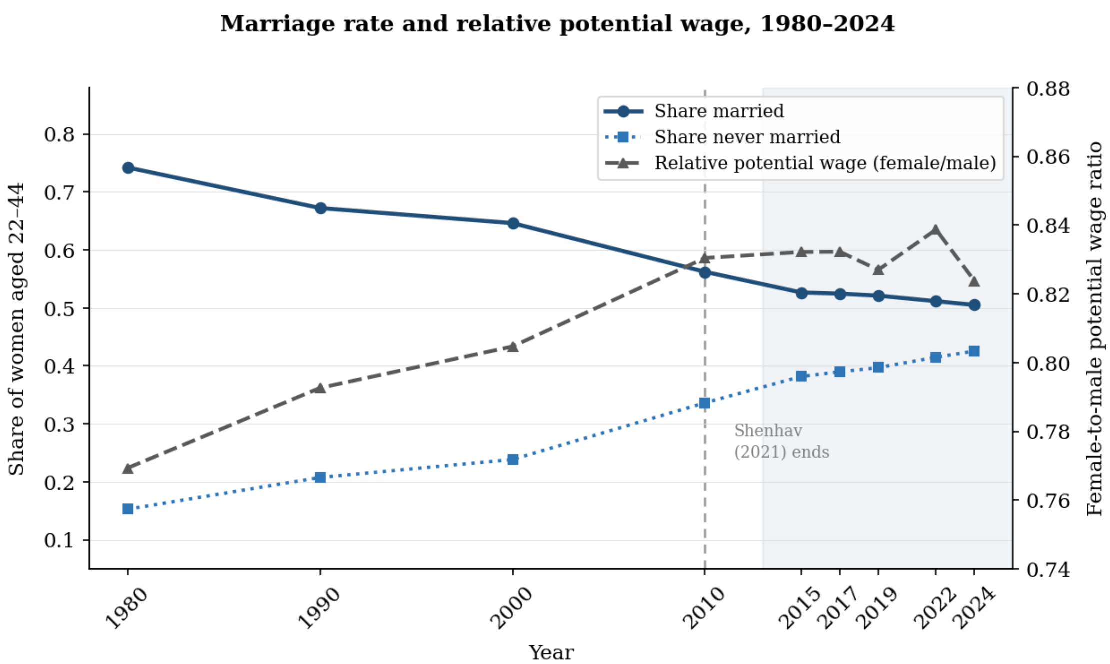

Figure 1: Marriage rate and relative potential wage, 1980 to 2024. The left axis shows the share married and the share never married among women aged 22 to 44, weighted by population across ACS markets. The right axis shows the female-to-male relative potential wage in levels, computed as the exponential of the mean log relative wage weighted by population, and therefore a ratio of geometric means. The vertical dashed line marks the end of the sample in Shenhav (2021), and the shaded band is the extension period.

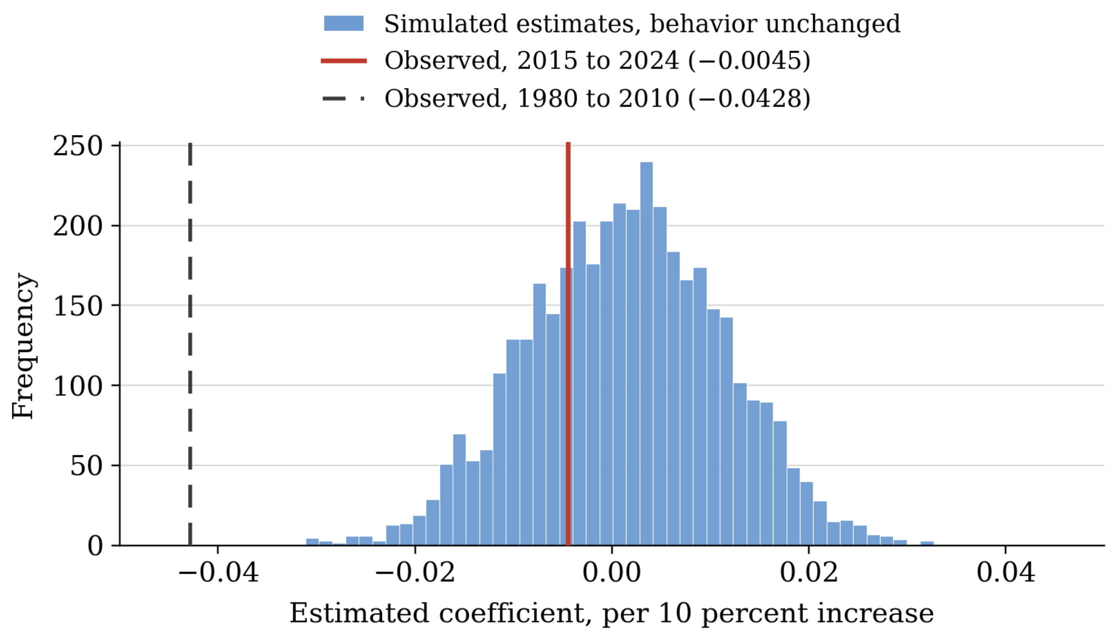

Figure 2: Simulated estimates of the marriage coefficient for 2015 to 2024 when behavior does not change and only the measure decouples. The simulation is calibrated to Table 2: the behavioral parameter is fixed at the value implied by the original period, $\beta^* = -0.0224$, and the attenuation factor is set to $\kappa = -0.066$, its value in the new waves. The residual variance is chosen so that the simulated standard error reproduces the one we report. Across 4,000 draws the simulated estimates have a mean of $+0.0014$ and 95 percent of them lie between -0.0184 and $+0.0209$. The estimate we obtain for 2015 to 2024 is an ordinary draw from this distribution. The estimate from the original period is not.